9.1.3 Fungible and non-fungible tokens

An ERC20 token is like currency, letting users buy certain utility or services by spending the tokens. There is an ERC20 token for paying for energy consumption in the Grid+ application, for example. One ERC20-compliant token of a kind (such as the REP token of Augur) is exchangeable with a token of the same kind, which means that it is a fungible token (FT).

A dollar bill can be exchanged with another dollar bill, so it is fungible. When a token represents an asset or a pet, like a real-world puppy, and it grows into a super dog that wins a world competition, this token value would appreciate enormously. Pokémon cards are another example. How about baseball cards and real estate? In these cases and many more practical examples, a given token value may appreciate or depreciate depending on many factors. This type of token is known as a *non-fungible* token (NFT). In this case, tokens are of the same kind, but they are not equal in value and so are not exchangeable for equal value.

> **DEFINITION** A fungible token (FT) is identical to every other token in value for the same class of tokens. One FT is equally exchangeable with any other within the given class.

> **DEFINITION** A non-fungible token (NFT) is a unique token within given token class. An NFT is not equal to any other NFT within the given class.

Figure 9.2 further illustrates the concepts of fungible and non-fungible.